

Missing Numbers in the Figure

Missing numbers are given in the form of figures in which some numbers are given. We have to find out the logic from one figure and put this logic to another figure and generate the answer.

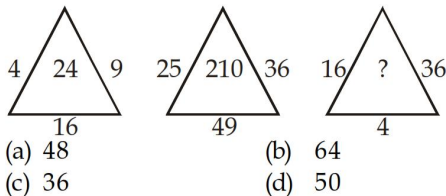
In such types of questions a figure consists of some (2, 3, 4 etc) parts or cells. Each cell has a number which has some logical relation with numbers in other cells and follows a certain rule. One has to identify the missing number, marked by the question mark "?"

There is no limit to the types of patterns, but the basic thing is to find the inter-connection between numbers. There are four basic operations in arithmetic :

Addition (+), Subtraction (-), Multiplication (×) and Division (÷). The missing number can be found with the manipulation of these basic operations. The candidate should require a lot of practice for such type of questions. It is better if he/she designs his/her own patterns and plays with numbers. This will help to enhance the imagination of the candidate.

Solved Examples

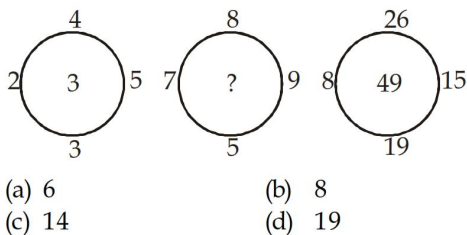
1.



Sol. (a); Clearly the digits or numbers along the side of triangles are square.

$$\begin{array}{lll} 2^2 = 4 & 5^2 = 25 & 4^2 = 16 \\ 3^2 = 9 & 6^2 = 36 & 6^2 = 36 \\ 4^2 = 16 & 7^2 = 49 & 2^2 = 4 \\ 2 \times 3 \times 4 = 24 & 5 \times 6 \times 7 = 210 & 4 \times 6 \times 2 = 48 \end{array}$$

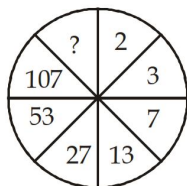
2.



Sol. (a); Here the pattern is first difference between right and left, bottom and top digits or number and then multiply the digits or numbers.

$$(4-3) \times (5-2) = 3; (8-5) \times (9-7) = 6 \\ (26-19) \times (15-8) = 49$$

3.

**Sol. (d);** $2 \times 2 - 1 = 3$

$7 \times 2 - 1 = 13$

$27 \times 2 - 1 = 53$

$107 \times 2 - 1 = 213$

$3 \times 2 + 1 = 7$

$13 \times 2 + 1 = 27$

$53 \times 2 + 1 = 107$

4.

9	15	13
7	30	14
2	?	3
8	5	9

(a) 5

(b) 6

(c) 9

(d) 25

Sol. (c); First, the sum of two number or digit then divided by lower digit or number and then answer such:

$$(9+7) \div 2 = 8, (13+14) \div 3 = 9, (15+30) \div$$

$$\boxed{9} = 5$$

5.

7	21	56
6	15	42
9	7	32
5	9	?

(a) 27

(b) 28

(c) 39

(d) 46

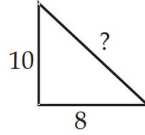
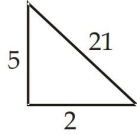
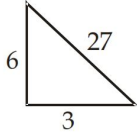
Sol. (b); Number at third place from left to right = $2 \times$ first number or digit + $2 \times$ second number or digit from left to right.

$$(7 \times 2) + (21 \times 2) = 56 \quad (6 \times 2) + (15 \times 2) = 42$$

$$(9 \times 2) + (7 \times 2) = 32 \quad (5 \times 2) + (9 \times 2) = 28$$

Practice Set

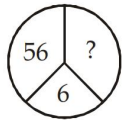
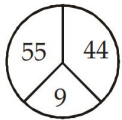
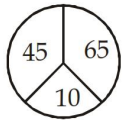
1.



- (a) 57
(c) 50

- (b) 54
(d) 60

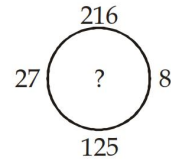
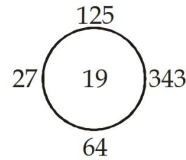
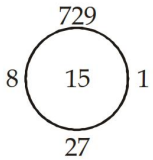
2.



- (a) 15
(c) 12

- (b) 10
(d) 17

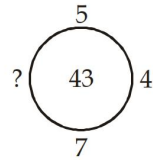
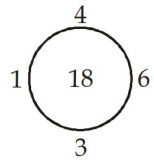
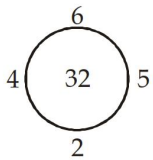
3.



- (a) 16
(c) 20

- (b) 18
(d) 29

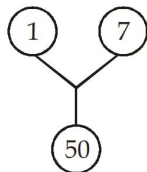
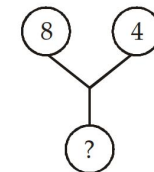
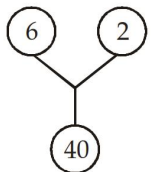
4.



- (a) 1
(c) 4

- (b) 2
(d) 5

5.



- (a) 80
(c) 40

- (b) 70
(d) 65

6.

5	4	7
6	10	8
4	11	?
60	220	56

- (a) 2
(c) 6

- (b) 4
(d) 10

7.

30	17	21
12	8	9
6	7	7
12	2	?

- (a) 4
(c) 11

- (b) 5
(d) 12

8.

49	36	64
25	16	100
4	9	121
14	?	29

- (a) 43
(c) 45

- (b) 69
(d) 13

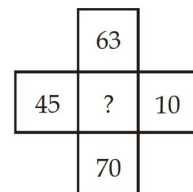
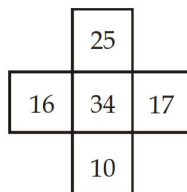
9.

9	6	7
8	12	5
3	2	?

- (a) 10
(c) 8

- (b) 11
(d) 15

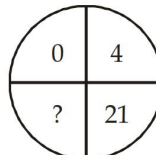
10.



- (a) 94
(c) 76

- (b) 95
(d) 10

11.



- (a) 120
(c) 140

- (b) 111
(d) 150

12. What will be at the place of x and y.

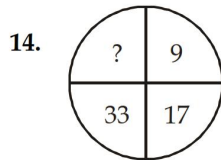
4B	36D	9F
5A	X	6E
6M	420	Y

- (a) 31C, 4P (b) 30C, 5P
(c) 30C, 7Q (d) 40D, 6M

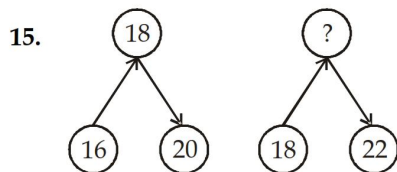
13.

2	7	9
7	3	4
9	8	?
126	168	216

- (a) 8 (b) 3
(c) 6 (d) 36



- (a) 60 (b) 68
(c) 55 (d) 65



- (a) 20 (b) 21
(c) 23 (d) 25

16.

9	4	8
6	5	2
2	2	4
12	3	?

- (a) 10 (b) 20
(c) 30 (d) 40

17.

2	14	28
3	9	27
4	?	40

- (a) 10 (b) 16
(c) 12 (d) 32

18.

81	36	25
49	100	36
9	64	16
139	200	?

- (a) 77 (b) 107
(c) 67 (d) 57

19.

169	64	81	30
625	?	49	50
1296	576	100	70

- (a) 324 (b) 289
(c) 441 (d) 361

20.

649	1084
253	?
143	482

- (a) 401 (b) 586
(c) 301 (d) 463

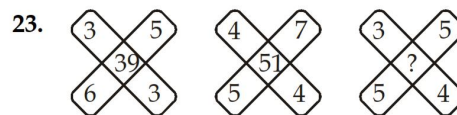
21. $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 6 & 10 & ? \end{pmatrix}$

- (a) 18 (b) 24
(c) 14 (d) 16

22.

1	2	3
4	5	6
7	8	9
27	38	?

- (a) 49 (b) 50
(c) 51 (d) 52



- (a) 35 (b) 37
(c) 45 (d) 47

24.

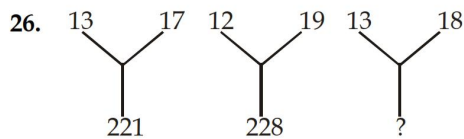
9	6	8
5	8	4
7	4	?
11	2	7

- (a) 4 (b) 7
(c) 3 (d) 6

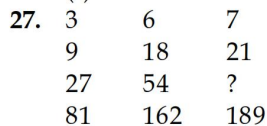
25. Find the missing number from the given responses :

4	9	17	6
20	5	8	9
7	23	9	9
?	9	4	19

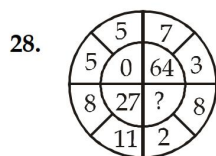
- (a) 27 (b) 29
(c) 22 (d) 26



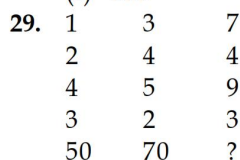
- (a) 31 (b) 229
(c) 234 (d) 312



- (a) 22 (b) 63
(c) 190 (d) 55



- (a) 216 (b) 125
(c) 100 (d) 144



- (a) 23 (b) 115
(c) 118 (d) 220

30. Find the missing number (?)

3	4	5
2	3	4
1	2	3
14	29	?

- (a) 50 (b) 30
(c) 40 (d) 32

31. Find the missing number from the given responses.

8	9	10
5	4	3
28	?	16
12	25	14

- (a) 28 (b) 11
(c) 32 (d) 18

32. Select the missing number from the given alternatives.

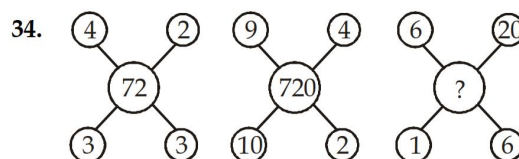
3	7	5
4	2	?
5	4	6
60	56	90

- (a) 9 (b) 3
(c) 4 (d) 8

33. The number of road accidents in a city is increasing month by month as given below. Find out the number of accidents in the month of June.

Month	Jan.	Feb.	March	April	May	June
No. of accidents	4	8	16	28	44	?

- (a) 64 (b) 48
(c) 52 (d) 40

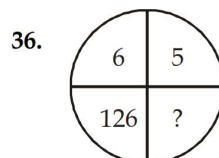


- (a) 72 (b) 720
(c) 7200 (d) 38

35. Find the missing number from the given responses :

5	6	12
4	3	4
2	3	?
18	27	96

- (a) 4 (b) 5
(c) 3 (d) 6

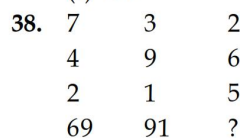


- (a) 127 (b) 31
(c) 217 (d) 328

37.

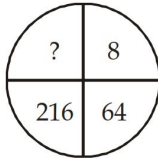
2	1	2
21	22	?
1	2	5
20	23	43

- (a) 40 (b) 48
(c) 50 (d) 36



- (a) 58 (b) 51
(c) 65 (d) 64

39.



- (a) 343 (b) 512
(c) 729 (d) 1000

40. The population of rats is increasing year after year in a village. Find out the missing population from the following information :

Years	1990	1991	1992	1993	1994	1995
Population	4	8	16	?	44	64

- (a) 22 (b) 32
(c) 28 (d) 34

Distinct Questions

41.

5	12	22
13	?	32
12	5	29

- (a) 16 (b) 32
(c) 4 (d) 6

42.

?	25200	5
600		5
120		10
	30	

- (a) 1800 (b) 400
(c) 3600 (d) 4500

43.

M	A	G
J	H	?
W	I	T

- (a) M (b) K
(c) A (d) L

44.

$$\begin{pmatrix} 2 & 3 & 1 \\ 1 & 2 & -1 \\ 3 & 4 & ? \end{pmatrix}$$

- (a) 5 (b) 2
(c) 1 (d) -2

45. Find the missing number from the given responses.

173	(24)	526
431	(18)	325
253	(?)	471

- (a) 22 (b) 42
(c) 30 (d) 06

46. What is the number missing from the target ?

5	9	15
16	29	?
49	89	147

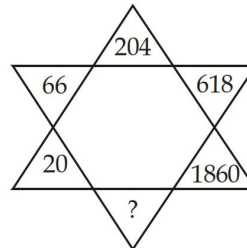
- (a) 45 (b) 48
(c) 51 (d) 54

47.

6	9	12
36	81	144
24	63	?

- (a) 120 (b) 80
(c) 94 (d) 102

48.



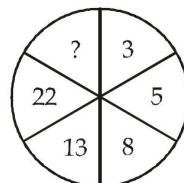
- (a) 5052 (b) 5586
(c) 5094 (d) 4860

49.

AZ	BY	CX
DW	EV	FU
GT	?	IR

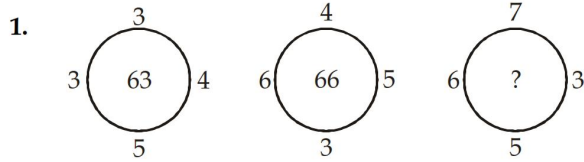
- (a) HR (b) HS
(c) HV (d) HU

50.

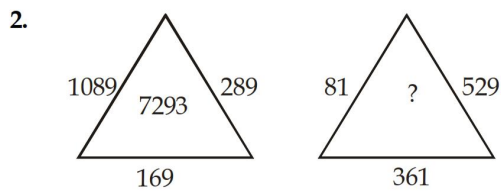


- (a) 1 (b) 26
(c) 39 (d) 45

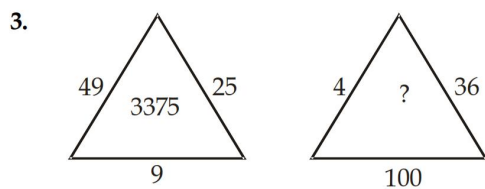
Previous year Questions (Memory Based)



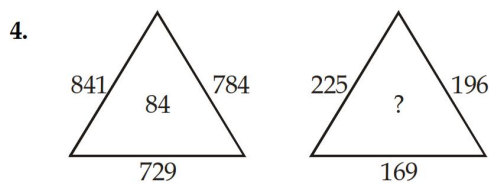
- (a) 57 (b) 53
(c) 105 (d) 111



- (a) 3646 (b) 3189
(c) 3399 (d) 3933



- (a) 2856 (b) 4268
(c) 5832 (d) 6464



- (a) 82 (b) 62
(c) 42 (d) 32

5.

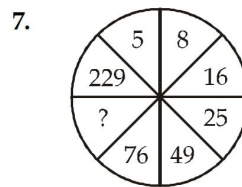
9	5	6
7	6	7
4	8	?
252	240	210

- (a) 4 (b) 5
(c) 6 (d) 3

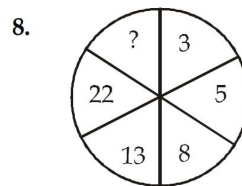
6.

5	17	23
7	8	2
9	15	5
42	80	?

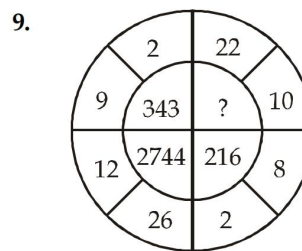
- (a) 50 (b) 60
(c) 70 (d) 98



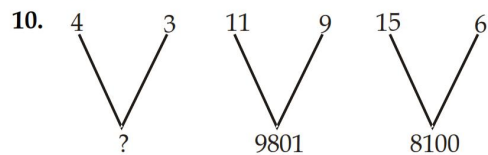
- (a) 148 (b) 150
(c) 125 (d) 53



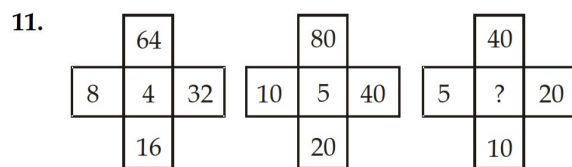
- (a) 1 (b) 26
(c) 39 (d) 45



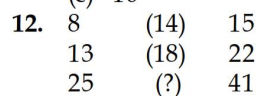
- (a) 1000 (b) 1728
(c) 878 (d) 560



- (a) 2250 (b) 144
(c) 11036 (d) 1216

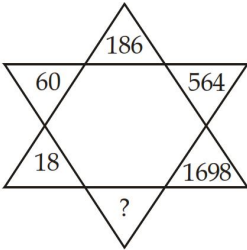


- (a) 0 (b) 2.5
(c) 10 (d) 20



- (a) 33 (b) 42
(c) 14 (d) 32

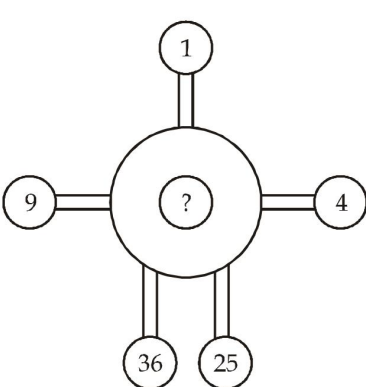
13. $\begin{array}{ccc} 2 & 2 & 256 \\ 3 & 2 & ? \\ 4 & 2 & 46656 \end{array}$
 (a) 2765 (b) 3125
 (c) 8796 (d) 3008
14. $6 \times 3 = 13$ $5 \times 20 = 96$
 $11 \times 7 = 67$ $19 \times 11 = ?$
 (a) 191 (b) 194
 (c) 207 (d) 209

15. 
 (a) 5052 (b) 5100
 (c) 5656 (d) 5510

16.

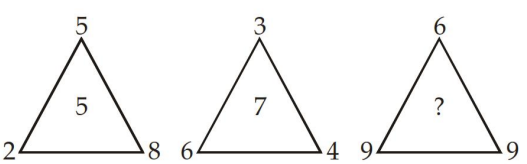
M	H	E
R	I	?
V	K	K

 (a) H (b) I
 (c) G (d) F

17. 

- (a) 15 (b) 27
 (c) 16 (d) 28
18. $94 + 16 = 42$
 $89 + 23 = 78$
 $63 + 45 = ?$
 (a) 18 (b) 28
 (c) 38 (d) 48

19. $\begin{array}{ccc} 49 & 100 & 64 \\ 9 & 36 & 4 \\ 81 & 1 & 25 \\ 19 & ? & 15 \end{array}$
 (a) 14 (b) 16
 (c) 17 (d) 18

20. 
 (a) 10 (b) 9
 (c) 12 (d) 11

21.

13	9	24
11	?	6
16	20	10

 (a) 19 (b) 16
 (c) 11 (d) 20

22.

2	4
256	16

3	1
1	81

5	4
256	?

 (a) 625 (b) 1225
 (c) 125 (d) 25

23.

1	3	7
2	4	4
4	5	9
3	2	3
50	70	?

 (a) 118 (b) 220
 (c) 23 (d) 115

24.

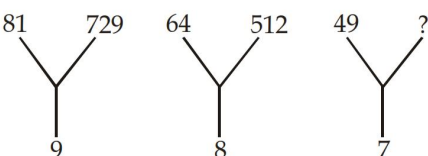
	18	
9	38	6
	17	

	11	
19	32	9
	11	

	6	
26	44	3
	15	

	12	
9	?	20
	8	

 (a) 31 (b) 40
 (c) 7 (d) 36

25. 
 (a) 444 (b) 515
 (c) 343 (d) 373

26.

8	17	33
12	5	29
10	13	?

- (a) 9 (b) 23
(c) 33 (d) 43

27.

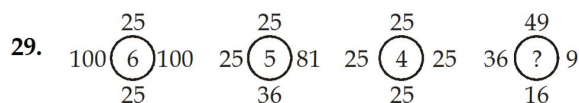
963	2	844
464	?	903

- (a) 1 (b) 2
(c) 3 (d) 4

28.

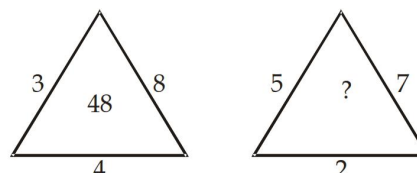
16	210	14
14	156	12
12	?	10

- (a) 90 (b) 100
(c) 110 (d) 120



- (a) 2 (b) 3
(c) 4 (d) 5

30.



- (a) 27 (b) 35
(c) 54 (d) 64

Practice Set Solutions

1. (b); In given sequence $(6 + 3) \times 3 = 27$, $(5 + 2) \times 3 = 21$, so $(10 + 8) \times 3 = 54$

2. (b); $(45 + 65) \div 11 = 10$, $(55 + 44) \div 11 = 9$ So, $\frac{56 + 10}{11} = 6$

3. (a); In given figure, outside present digits are perfect cube of number add all cube root of given number, so $\sqrt[3]{216} + \sqrt[3]{8} + \sqrt[3]{27} + \sqrt[3]{125} = 16$.

4. (b); Multiplication of opposite values and then their addition.

$$5 \times 7 + 4x = 43, \text{ so, } x = 2$$

5. (a); Given lowest digit is sum of square of given upper digits. So $8^2 + 4^2 = 64 + 16 = 80$

6. (a); $\frac{5 \times 4 \times 6}{2} = 60$, $\frac{4 \times 10 \times 11}{2} = 220$ then $\frac{7 \times 8 \times x}{2} = 56$ so x is 2.

7. (b); From downwards, $(12 + 6 + 12) = 30$. $(8 + 7 + 2) = 17$ so, $x + 9 + 7 = 21$, so $x = 5$.

8. (d); $\sqrt{49} + \sqrt{25} + \sqrt{4} \Rightarrow 7 + 5 + 2 = 14$, so $\sqrt{36} + \sqrt{16} + \sqrt{9} = 13$.

9. (c); The sum of every column is 20 then value will be 8.

10. (a); The exact middle digit is just half of sum of all outside digit.

11. (b); $0 \times 3 + 4 = 4$, $4 \times 4 + 5 = 21$, so $21 \times 5 + 6 = 111$

12. (c); In middle column in alphabets there is 2 place value difference and digit is multiplication of both first and third digit.

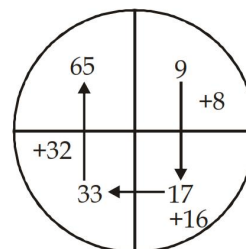
13. (c); Columnwise

$$\text{I. } 2 \times 7 \times 9 = 126 \quad \text{II. } 7 \times 3 \times 8 = 168$$

$$\text{III. } 9 \times 4 \times ? = 216$$

$$\therefore ? = \frac{216}{9 \times 4} = 6$$

14. (d);



15. (a); 8×2 , 9×2 , 10×2 in first diagram same as 9×2 , 10×2 , 11×2 .

16. (c); $9 \times 2 - 6 = 12$, $4 \times 2 - 5 = 3$ so $8 \times 4 - 2 = 30$.

17. (a); $2 \times 14 = 28$, $3 \times 9 = 27$, so $4 \times x = 40$, so x is 10.

18. (a); Fourth last term of every column is sum of all above values, so $25 + 36 + 16 = 77$.

19. (a); First Row

$$\Rightarrow \sqrt{169} + \sqrt{64} + \sqrt{81} \\ = 13 + 8 + 9 = 30$$

Second Row

$$\Rightarrow 25 + \sqrt{?} + \sqrt{49} = 50$$

$$\Rightarrow \sqrt{?} = 50 - 32 = 18$$

$$\therefore (?) = (18)^2 = 324$$

Third Row

$$\Rightarrow \sqrt{1296} + \sqrt{576} + \sqrt{100} \\ = 36 + 24 + 10 = 70$$

20. (c); $\frac{649 - 143}{2} = (253)$, so $\frac{1084 - 482}{2} = \frac{602}{2} = 301$

21. (c); Here we get the lowermost term by adding the two uppermost terms and doubling it.

$$\text{As, } (1 + 2) \times 2 = 6 \quad (3 + 2) \times 2 = 10,$$

Similarly,

$$(3 + 4) \times 2 = 14.$$

22. (c); First Column $= 4 \times 7 - 1 = 28 - 1 = 27$
 Second Column $= 5 \times 8 - 2 = 40 - 2 = 38$
 Third Column $= 6 \times 9 - 3 = 54 - 3 = 51$

23. (b); $5 \times 6 + 3 \times 3 = 39$
 and, $7 \times 5 + 4 \times 4 = 51$
 $\therefore 5 \times 5 + 3 \times 4 = 37$

24. (c); $9 + 7 - 5 = 11$
 and, $6 + 4 - 8 = 2$
 $\therefore 8 + ? - 4 = 7 \Rightarrow ? = 3$

25. (c); $4 + 9 + 17 + 6 = 36$
 $20 + 5 + 8 + 9 = 42$
 $7 + 23 + 9 + 9 = 48$
 $9 + 4 + 19 + ? = 54$

$\rightarrow 54 - 32 = 22$

26. (c); $13 \times 17 = 221$
 $12 \times 19 = 228$
 $13 \times 18 = 234$

27. (b); $3 \times 3 = 9, 9 \times 3 = 27, 27 \times 3 = 81$
 $6 \times 3 = 18, 18 \times 3 = 54, 54 \times 3 = 162$
 $7 \times 3 = 21, 21 \times 3 = 63, 63 \times 3 = 189$

28. (a); $5 - 5 = 0 \Rightarrow 0^3 = 0$
 $= 7 - 3 = 4, \Rightarrow 4^3 = 64$
 $11 - 8 = 3 \Rightarrow 3^3 = 27$
 $8 - 2 = 6 \Rightarrow 6^3 = 216$

29. (b); $1 + 2 + 4 + 3 = 10;$
 $10 \times 5 = 50$
 $3 + 4 + 5 + 2 = 14;$
 $14 \times 5 = 70$
 $7 + 4 + 9 + 3 = 23;$
 $23 \times 5 = \boxed{115}$

30. (a); $(3)^2 + (2)^2 + (1)^2 = 9 + 4 + 1 = 14$
 $(4)^2 + (3)^2 + (2)^2 = 16 + 9 + 4 = 29$
 Similarly
 $(5)^2 + (4)^2 + (3)^2 = 25 + 16 + 9 = 50$

31. (b); $8 \times 5 - 28 = 40 - 28 = 12$
 $10 \times 3 - 16 = 30 - 16 = 14$
 $9 \times 4 = ? = 25$

$\Rightarrow 36 - ? = 25$
 $\therefore ? = 36 - 25 = 11$

32. (b); $3 \times 4 \times 5 = 60 \quad 7 \times 2 \times 4 = 56$
 $5 \times ? \times 6 = 90$

$\therefore ? = \frac{90}{5 \times 6} = 3$

33. (a); $4 + 4 = 8, 8 + 8 = 16, 16 + 12 = 28$
 $28 + 16 = 44, 44 + 20 = 64$

34. (b); $4 \times 2 \times 3 \times 3 = 72$
 $9 \times 4 \times 2 \times 10 = 720$

$6 \times 20 \times 1 \times 6 = \boxed{720}$

35. (d); $5 + 4 = 9$ and $9 \times 2 = 18$
 $6 + 3 = 9$ and $9 \times 3 = 27$

$12 + 4 = 16$ and $? = \frac{96}{16} = \boxed{6}$

36. (c); $(5)^3 + 1 = 125 + 1 = 126$
 $(6)^3 + 1 = 216 + 1 = 217$

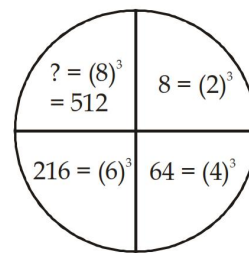
37. (a); $(21 + 1) - 2 = 22 - 2 = 20$
 $(22 + 2) - 1 = 24 - 1 = 23$
 $(? + 5) - 2 = 43$

$\Rightarrow ? = (43 + 2) - 5$

$\Rightarrow ? = 45 - 5 = 40$

38. (c); Columnwise
 First Column $(7)^2 + (4)^2 + (2)^2 = 49 + 16 + 4 = 69$
 Second Column $(3)^2 + (9)^2 + (1)^2 = 9 + 81 + 1 = 91$
 Third Column $(2)^2 + (6)^2 + (5)^2 = 4 + 36 + 25 = 65$

39. (b);



$? = (8)^3 = 512 \quad 8 = (2)^3$
 $216 = (6)^3 \quad 64 = (4)^3$

40. (a); $4 + 4 = 8 \quad 8 + 8 = 16$
 $16 + 12 = 28 \quad 28 + 16 = 44 \quad 44 + 20 = 64$

Distinct Solutions

41. (d); $(5 \times 2) + 12 = 22, (12 \times 2) + 5 = 29$, so $(13 \times 2) + x = 32$, so $x = 6$.

42. (c); From top digit 5 in clockwise direction multiplication of natural number 1, 2, 3, 4, 5. So, $600 \times 6 = 3600$.

43. (a); $M(13) + J(10) = W(23), (A)(1) + H(8) = I(9)$
 So, $G(7) + x = 20 \Rightarrow 20 - 7 = 13$ (M) so ans. is M.

44. (a); Here the number series is as follows-
 $2^2 - 3 = 1$
 $1^2 - 2 = -1$
 $\therefore 3^2 - 4 = 5$.

45. (a); $1 + 7 + 3 + 5 + 2 + 6 = 24$
 $4 + 3 + 1 + 3 + 2 + 5 = 18$

Therefore,
 $2 + 5 + 3 + 4 + 7 + 1 = 22$
 46. (b); $5 \times 3 + 1 = 16$ $16 \times 3 + 1 = 49$
 $9 \times 3 + 2 = 29$ $29 \times 3 + 2 = 89$

Therefore,
 $15 \times 3 + 3 = 48$ $48 \times 3 + 3 = 147$

47. (a); Columnwise
 $6 \times 6 = 36$ $6 \times (6 - 2) = 6 \times 4 = 24$
 $9 \times 9 = 81$ $9 \times (9 - 2) = 9 \times 7 = 63$
 $12 \times 12 = 144$
 $12 \times (12 - 2) = 12 \times 10 = 120$

48. (b); $20 \times 3 + 6 = 60 + 6 = 66$
 $66 \times 3 + 6 = 198 + 6 = 204$

$204 \times 3 + 6 = 612 + 6 = 618$
 $618 \times 3 + 6 = 1854 + 6 = 1860$

$1860 \times 3 + 6 = 5580 + 6 = \boxed{5586}$

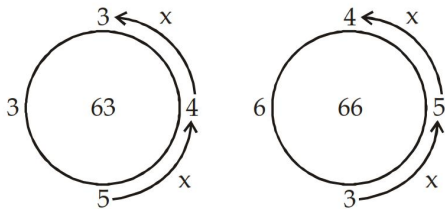
49. (b); In each sector pair of opposite letters are given.

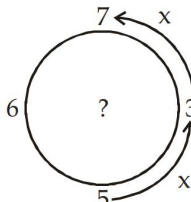
$G \xrightarrow{+1} H$

The opposite letter of H is S.

50. (c); $3 \times 2 - 1 = 6 - 1 = 5$
 $5 \times 2 - 2 = 10 - 2 = 8$
 $8 \times 2 - 3 = 16 - 3 = 13$
 $13 \times 2 - 4 = 26 - 4 = 22$
 $22 \times 2 - 5 = 44 - 5 = 39$

Previous year Solutions (Memory Based)

1. (d); 
 $5 \times 4 \times 3 = 60 + 3 = 63$ $3 \times 5 \times 4 = 60 + 6 = 66$


 $5 \times 3 \times 7 = 105 + 6 = 111$

2. (d); $\sqrt{1089} \times \sqrt{289} \times \sqrt{169} = 33 \times 17 \times 13 = 7293$
 $\sqrt{81} \times \sqrt{529} \times \sqrt{361} = 9 \times 23 \times 19 = 3933$

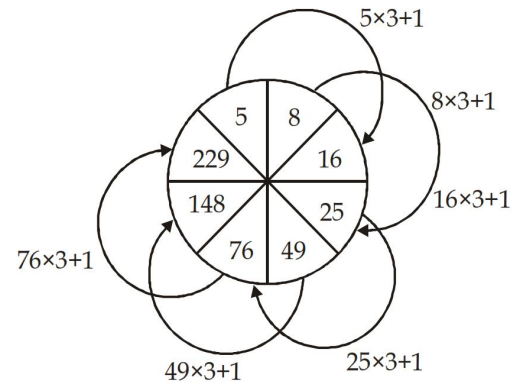
3. (c); $(\sqrt{49} + \sqrt{9} + \sqrt{25})^3 = (7 + 3 + 5)^3 = (15)^3 = 3375$
 $(\sqrt{4} + \sqrt{100} + \sqrt{36})^3 = (2 + 10 + 6)^3 = (18)^3 = 5832$

4. (c); $\sqrt{841} + \sqrt{784} + \sqrt{729} = 29 + 28 + 27 = 84$
 $\sqrt{225} + \sqrt{196} + \sqrt{169} = 15 + 14 + 13 = 42$

5. (b); $9 \times 7 \times 4 = 252$; $5 \times 6 \times 8 = 240$; $6 \times 7 \times \boxed{5} = 210$

6. (b); $5 + 7 + 9 = 21 \times 2 = 42$
 $17 + 8 + 15 = 40 \times 2 = 80$
 $23 + 2 + 5 = 30 \times 2 = 60$

7. (a);



8. (c); $3 \times 2 - 1 = 5$ $5 \times 2 - 2 = 8$ $8 \times 2 - 3 = 13$
 $13 \times 2 - 4 = 22$ $22 \times 2 - 5 = \boxed{39}$

9. (b); $9 - 2 = 7^3 = 343$ $8 - 2 = 6^3 = 216$
 $26 - 12 = 14^3 = 2744$ $22 - 10 = 12^3 = 1728$

10. (b); $11 \times 9 = 99^2 = 9801$ $15 \times 6 = 90^2 = 8100$
 $4 \times 3 = 12^2 = 144$

11. (b); All numbers are divided by 4 in 1st figure. All number are divided by 5 in 2nd figure. In the 3rd figure, all numbers will be divided by 2.5

12. (d); $8 + \frac{14}{2} = 15$ $13 + \frac{18}{2} = 22$ $25 + \frac{\boxed{32}}{2} = 41$

13. (b); $(2 + 2) = 4 \times 4 \times 4 \times 4 = 256$
 $(3 + 2) = 5 \times 5 \times 5 \times 5 = 3125$
 $(4 + 2) = 6 \times 6 \times 6 \times 6 \times 6 = 46656$

14. (a); $6 \times 3 \rightarrow 18 - (6 - 1) = 13$
 $5 \times 20 \rightarrow 100 - (5 - 1) = 96$
 $11 \times 7 \rightarrow 77 - (11 - 1) = 67$
 $19 \times 11 \rightarrow 209 - (19 - 1) = 191$

15. (b); $18 \times 3 = 54 + 6 \quad 60$
 $60 \times 3 = 180 + 6 \rightarrow 186$
 $186 \times 3 = 558 + 6 \rightarrow 564$
 $564 \times 3 = 1692 + 6 = 1698$
 $1698 \times 3 = 5094 + 6 = 5100$

16. (b); $M(13) = H(8) + E(5)$
 $V(22) = K(11) + K(11)$
 $R(18) = I(9) + I(9)$

17. (c); The square of digits are in series
 $1^2 \quad 2^2 \quad 3^2 \quad 4^2 \quad 5^2 \quad 6^2$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $1 \quad 4 \quad 9 \quad 16 \quad 25 \quad 36$

18. (c); $(9 \times 4) + (1 \times 6) = 36 + 6 = 42$
 $(8 \times 9) + (2 \times 3) = 72 + 6 = 78$
 $(6 \times 3) + (4 \times 5) = 18 + 20 = \boxed{38}$

19. (c); $\sqrt{49} + \sqrt{9} + \sqrt{81} = 7 + 3 + 9 = 19$
 $\sqrt{64} + \sqrt{4} + \sqrt{25} = 8 + 2 + 5 = 15$
 $\sqrt{100} + \sqrt{1} + \sqrt{36} = 10 + 1 + 6 = 17$

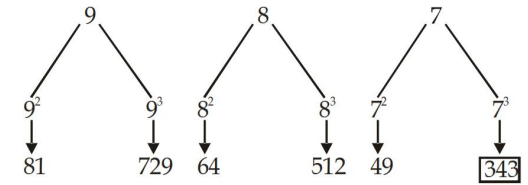
20. (c); $(2 + 8) = (5 + 5) \quad (3 + 7) = (6 + 4)$
 $(9 + 9) = (12 + 6)$

21. (c); $13 + 11 + 16 = 40 \quad 24 + 6 + 10 = 40$
 $9 + ? + 20 = 40 \Rightarrow ? = 40 - 29 = 11$

22. (a); $2^4 = 16 ; 4^4 = 256$
 $1^4 = 1 ; 3^4 = 81$
 $4^4 = 256 ; 5^4 = 625$

23. (d); $(1 + 2 + 4 + 3) \times 5 = 50$
 $(3 + 4 + 5 + 2) \times 5 = 70$
 $(7 + 4 + 9 + 3) \times 5 = 115$

24. (a); $(17 + 18) + (9 - 6) = 35 + 3 = 38$
 $(11 + 11) + (19 - 9) = 22 + 10 = 32$
 $(15 + 6) + (26 - 3) = 21 + 23 = 44$
 $(12 + 8) + (9 - 20) = 20 + 11 = \boxed{31}$

25. (c); 

26. (c); $8 \times 2 + 17 = 33 \quad 12 \times 2 + 5 = 29$
 $10 \times 2 + 13 = 33$

27. (b); $(9 + 6 + 3) - (8 + 4 + 4) = 2$
 $(4 + 6 + 4) - (9 + 0 + 3) = 2$

28. (c); $16 \times 14 - 14 = 210 \quad 14 \times 12 - 12 = 156$
 $12 \times 10 - 10 = 110$

29. (c); $\frac{\sqrt{25} + \sqrt{100} + \sqrt{25} + \sqrt{100}}{5} = 6$

$$\frac{\sqrt{25} + \sqrt{81} + \sqrt{36} + \sqrt{25}}{5} = 5$$

$$\frac{\sqrt{25} + \sqrt{25} + \sqrt{25} + \sqrt{25}}{5} = 4$$

$$\frac{\sqrt{49} + \sqrt{9} + \sqrt{16} + \sqrt{36}}{5} = 4$$

30. (b); $(3 \times 8 \times 4) \div 2 = 48$
 $(5 \times 7 \times 2) \div 2 = 35$

